

Arnav Khamparia

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Education

B.Tech in Artificial Intelligence and Machine Learning
Madhav Institute of Technology and Science
Relevant Coursework: Advance NLP with Transformers , Machine Learning, Scalable ML Infrastructure, Data Structures.

2021 – 2025
Gwalior, India

Professional Experience

AIML Intern
OTO Capital

Nov 2025 – Present
Bengaluru, India

- Engineered prompt systems for a production AI Voice Agent powering real-time borrower conversation built on Google Gemini Live API for ultra-low latency, moving beyond traditional STT-TTS pipelines.
- Designed an AI-driven outreach orchestration engine that autonomously decides the best channel (AI Call / WhatsApp / SMS / Manual) per borrower based on behavior signals functioning as a self-deciding rulebook.
- Built a disposition analysis system that processes call recordings, extracts structured insights, and feeds them back into the AI, creating a self-improving feedback loop with no human intervention.
- Operated at the intersection of ML engineering and product, owning end-to-end decisions on how the AI behaves, escalates, and learns, reducing friction across the entire collections pipeline.
- Collaborated with senior engineers and product teams to debug, deploy, and continuously improve reliable AI systems in a fast-moving fintech environment.

Machine Learning Intern
Vivada Tech

May 2024 – Aug 2024
Chennai, India

- Developed and optimized machine learning models using prompt engineering, improving SDK performance by 20% for AI evaluation and observability tools.
- Designed and integrated REST APIs and SDKs, reducing model deployment time by 15% and enabling seamless interaction with cloud-based ML pipelines.
- Authored technical documentation, increasing team onboarding efficiency by 30%.

Projects

Generative AI SEO Content Augmenter

Jul 2025

- Engineered a Generative AI-powered web application using LLM, Vite, and RAG to perform competitive SERP analysis and identify key content gaps for SEO optimization.
- Implemented a core feature leveraging AI to strategically augment user content by generating data-driven FAQs, Myth vs. Fact tables, and SEO-optimized FAQPage Schema markup.
- Designed and developed a responsive, intuitive user interface (UI) that streamlines the content strategy workflow for marketers, aimed at reducing manual research time and boosting content performance.

Skills

Programming: Python, SQL

Machine Learning: TensorFlow, PyTorch, Scikit-learn, LangChain, LangGraph, Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Transformer Architectures (BERT, GPT, T5), Prompt Engineering.

Tools and Platforms: Git, Streamlit, REST APIs, Postman, AWS

Domains: Deep Learning, Generative AI, Natural Language Processing (NLP), Computer Vision, Data Structures & Algorithms.

Publications

Liver Tumor Segmentation with U-Net, V-Net and AH-Net Using MONAI

Springer Book Series

Co-authored a study comparing U-Net, V-Net, and AH-Net for liver CT scan segmentation, achieving a top Dice score of 0.93 for volumetric analysis, contributing to improved diagnostic accuracy.

Accolades

MITS Alumni Community
Content Coordinator
Developed content strategies for newsletters and social media, increasing alumni engagement by 40% through targeted campaigns and event highlights.

Artificial Intelligence Club
Technical Lead
Led 10+ team projects and workshops on AI/ML, mentoring 20+ students in Python and TensorFlow, resulting in successful project deployments.